Fernando Lucatelli Nunes

Assistant Professor, University of Coimbra

arXiv preprints: `https://arxiv.org/a/lucatellinunes_f_1.html`

Google Scholar

Scopus ID 57188838707

Mathematics Genealogy

Webpage: https://flnlucatelli.github.io/ **E-mail:** fernandolucatellinunes@gmail.com

Assistant Professor at the Department of Mathematics of the University of Coimbra. In mathematics, I work on the theory and applications of generalised categorical structures, often through two- and three-dimensional universal algebra, with connections to proof theory, algebraic topology, universal algebra, logic, and related fields. In computer science, I study programming language semantics and correctness-by-construction approaches to the design and implementation of programming languages. More broadly, I regard structural and logical methods not as external tools, but as foundations for sound design: the aim is to develop coherent theories that guide implementation and yield systems that are reliable, transparent, and conceptually well-founded.

ACADEMIC APPOINTMENTS

2026 – present	**Assistant Professor**, Department of Mathematics, University of Coimbra, Portugal.
2025 – 2026	**Research Fellow**, School of Computer Science, University of Birmingham, United Kingdom.
2025	**Postdoctoral Fellow**, Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2023	**Postdoctoral Fellow**, York University, Canada.
2023	**Fields Research Fellow**, Fields Institute for Research in Mathematical Sciences, Canada.
2022	**Oberwolfach Leibniz Fellow**, Mathematisches Forschungsinstitut Oberwolfach, Germany.
2018 – 2021	**Research Fellow**, University of Coimbra, Portugal.
2018 – 2019	**Invited Assistant Professor**, University of Coimbra, Portugal.
2018	**Postdoctoral Fellow**, Centre for Mathematics, University of Coimbra, Portugal.

RESEARCH GROUPS AND AFFILIATIONS

2026 – present	**Integrated Researcher**, Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2025 – 2026	**Research Fellow**, Theory of Computation, School of Computer Science, University of Birmingham, United Kingdom.
2020 – 2026	**Researcher in Programming Language Semantics**, Software Technology Group, Utrecht University, The Netherlands.
2020 – 2024	**Research Collaborator**, Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2018 – 2020	**Postdoctoral Researcher, and Research Fellow**, Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2013 – 2018	**PhD candidate**, Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2013 – present	**Research Member**, Coimbra Category Theory Group, University of Coimbra, Portugal.

VISITING ACADEMIC APPOINTMENTS

2026	**Visiting Researcher**, Utrecht University.
2025	**Visiting Researcher**, IMPA, Rio de Janeiro, Brazil.
2018, 2023, 2024, 2025	**Visiting Researcher**, UCLouvain, Belgium.
2023	**Visiting Researcher**, York University, Canada.
2019, 2025	**Visiting Researcher**, University of Cape Town, South Africa.
2019, 2025	**Visiting Researcher**, Stellenbosch University, South Africa.

Education

PhD IN MATHEMATICS (AWARDED 2018)

10/2012 – 01/2018

University of Coimbra, PhD in Mathematics

- Thesis: *Pseudomonads and Descent* (awarded *summa cum laude*).
- Supervisor: Prof. Maria Manuel Clementino.
- PhD awarded in Jan/2018
- Examination Committee: Prof. Martin Hyland, Prof. Marco Mackaay, Prof. Lurdes Sousa, and Prof. Peter Beier Gothen.
- Main contributions: Biadjoint triangle theorems; freely generated categorical structures; theories of presentation and coherence for low- and higher-dimensional categorical structures; Grothendieck descent theory; categorical Galois theories; low- and higher-dimensional weak limits.

PhD IN COMPUTER SCIENCE (to be awarded in September, 2026)

11/2020 – 11/2025

Utrecht University, PhD in Computer Science

- Thesis: *Freely Generated Categorical Structures and Automatic Differentiation.*
- Supervisors: Prof. Gabriele Keller and Dr. Matthijs Vákár.
- PhD to be awarded Sep/2026
- Examination Committee: Prof. Ieke Moerdijk, Prof. Bart Jacobs, Prof. Sam Staton, Prof. Robin Cockett, and Prof. Patricia Johann.
- Main contributions: Denotational semantics of programming languages; automatic differentiation; principled program transformations; recursion and iteration; categorical semantics; logical relations; type systems and dependent types; Grothendieck constructions; extensive categories.

Research

RESEARCH INTERESTS

Expertise and research interests: In mathematics, I work on the theory and applications of low- and higher-dimensional generalised categorical structures, often developed through two- and three-dimensional universal algebra. This programme connects naturally with areas such as proof theory, the geometry of manifolds, algebraic topology, universal algebra, logic, and related fields.
In computer science, I work in programming language semantics, formal methods, and software technology, with particular emphasis on correctness-by-construction approaches to the design and implementation of programming languages. My research combines rigorous semantic analysis with questions arising from the engineering of modern programming paradigms and software systems.

SELECTED PUBLICATIONS

Lucatelli Nunes, F., Vákár, M.: **Monoidal closure of Grothendieck constructions via Σ-tractable monoidal structures and Dialectica formulas.** Theory Appl. Categ. 44, 1153–1217. 2025. (*65 pages*).
Lucatelli Nunes, F., Prezado, R., Vákár, M.: **Free extensivity via distributivity.** Portugaliae Mathematica, 82, no. 1/2, pp. 177–204. 2025. (*28 pages*).
Prezado, R., Lucatelli Nunes, F.: **Generalized multicategories: change-of-base, embedding, and descent.** Appl. Categ. Struct. Volume 32, 35. 2024. (*89 pages*).
Clementino, M.M., Lucatelli Nunes, F.: **Lax comma 2-categories and admissible 2-functors.** Theory Appl. Categ. 40, 180–226. 2024. (*47 pages*).
Clementino, M.M., Lucatelli Nunes, F., Prezado, R.: **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent.** Theory Appl. Categ. 41, 516–530. 2024. (*15 pages*).
Lucatelli Nunes, F., Vákár, M.: **Automatic differentiation for ML-family languages: Correctness via logical relations.** Math. Struct. Comput. Sci. 34, No. 8, 747–806. 2024. (60 pages).
Prezado, R., Lucatelli Nunes, F.: **Descent for internal multicategory functors.** Appl Categ. Struct 31, 11. 2023. (*18 pages*).
Clementino, M.M., Lucatelli Nunes, F.: **Lax comma categories of ordered sets.** Quaest. Math. 46, Part Suppl. 1, 145–159. 2023. (*14 pages*).
Lucatelli Nunes, F., Vákár, M.: **CHAD for Expressive Total Languages.** Math. Struct. Comput. Sci. 33, No. 4–5, 311–426. 2023. (*116 pages*).
Lucatelli Nunes, F., Prezado, R., Sousa, L.: **Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations.** Bull. Belg. Math. Soc. 30, No. 1, 130–139. 2023. (*10 pages*).
Lucatelli Nunes, F.: **Semantic factorization and descent.** Appl. Categ. Struct. 30, No. 6, 1393-1433. 2022. (*41 pages*).
Lucatelli Nunes, F., Sousa, L.: **On lax epimorphisms and the associated factorization.** J. Pure Appl. Algebra 226, No. 12, 27 p. 2022. (*27 pages*).
Lucatelli Nunes, F.: **Descent data and absolute Kan extensions.** Theory Appl. Categ. 37, 530–561. 2021. (*32 pages*).
Lucatelli Nunes, F.: **Pseudoalgebras and non-canonical isomorphisms.** Appl. Categ. Struct. 27, No. 1, 55–63. 2019. (*9 pages*).
Lucatelli Nunes, F.: **On lifting of biadjoints and lax algebras.** Categ. Gen. Algebr. Struct. Appl. 9, No. 1, 29–58. 2018. (*30 pages*).
Lucatelli Nunes, F.: **Pseudo-Kan Extensions and Descent Theory.** Theory Appl. Categ. 33, 390–444. 2018. (*35 pages*).
Lucatelli Nunes, F.: **On biadjoint triangles.** Theory Appl. Categ. 31, 217–256. 2016. (*40 pages*).

ACCEPTED FOR PUBLICATION

Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Unraveling the iterative CHAD.** arXiv:2505.15002. 2025. (*57 pages*). (Mathematical Structures in Computer Science.)
Lucatelli Nunes, F., Vákár, M.: **Free doubly-infinitary distributive categories are cartesian closed.** arXiv:2403.10447. 2024. (*16 pages*). (Applied Categ. Struct.)

PREPRINTS (UNDER REVIEW)

Lucatelli Nunes, F., Prezado, R.: **Functors Preserving Effective Descent Morphisms.** arXiv:2410.22876. 2024. (*17 pages*).
Lucatelli Nunes, F.: **Freely generated n-categories, coinserters and presentations of low dimensional categories.** arXiv:1704.04474. 2017. (*56 pages*).

SELECTED MANUSCRIPTS IN PREPARATION

Lucatelli Nunes, F., and Tholen, W.: **Spanning the tale of Monads and Descent.** Manuscript in preparation
Lucatelli Nunes, F., and Clementino, M.: **On lax comma 2-categories.** Manuscript in preparation
Lucatelli Nunes, F., Plotkin, G., and Vákár, M.: **Backprop to the Future: CHAD for Recursion.** Manuscript in preparation
Lucatelli Nunes, F., Goncharov, S., Vákár, M., and Van der Linden, T.: **Symmetric Infinitary Monoidal Categories.** Manuscript in preparation
Lucatelli Nunes, F., and Levy, P.B.: **In-place strictification.** Manuscript in preparation

OTHER PUBLICATIONS

Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Homomorphic Reverse Differentiation of Iteration.** LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024). 2024. (*2 pages*).
Lucatelli Nunes, F., Vákár, M.: **Logical Relations for Partial Features and Automatic Differentiation Correctness.** arXiv:2210.08530. 2022. and Oberwolfach Preprints, OWP-2023-09. 2023. (*21 pages*).
Lucatelli Nunes, F.: **Freely generated categorical structures and automatic differentiation.** Computer Science PhD Thesis, Utrecht University. 2025. (*380 pages*).
Lucatelli Nunes, F.: **Pseudomonads and Descent.** Mathematics PhD Thesis, University of Coimbra. 2017. (*209 pages*).
Lucatelli Nunes, F.: **Espaços não reversíveis.** (Non-reversible spaces). Revista Matemática Universitária, v. 48, n. 49, p. 22-26. 2012. (*5 pages*).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

Lucatelli Nunes, F.: **Basic Concepts of Topology for Topological Dynamics (in Portuguese).** 2009. (*116 pages*).
Lucatelli Nunes, F.: **Basic Topological Dynamics and Basic Applications to Number Theory (in Portuguese).** 2009. (*57 pages*).

Grants, Fellowships, and Funded Projects

SELECTED RESEARCH FELLOWSHIPS (PI)

2022 — **Oberwolfach Leibniz Fellowship**, Grothendieck Descent Theory, Galois Theory, and Presentations of Categorical Structures

- PI: **Fernando Lucatelli Nunes**.
- Participants: Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, Matthijs Vákár.

2023 — **Fields Research Fellowship**, Grothendieck Descent Theory, Fibrations, and Factorizations

- PI: **Fernando Lucatelli Nunes**.
- Participants: Walter Tholen.

SELECTED PARTICIPATION IN RESEARCH PROJECTS AND GRANTS

2025 – present — **ERC Starting Grant FoRECAST**, Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy

- PI/Grant holder: Matthijs Vákár.
- My role: Member of the formal methods team, contributing to the design of principled algorithms grounded in categorical structures, and involved in the (co-)supervision of PhD students.

2025 – 2026 — **DFG Project HOMBRe**, Higher-Order Monad-based Programming and Reasoning

- PI/Grant holder: Sergey Goncharov.
- My role: Postdoctoral Fellow, contributing to the core research themes of the project.

2021 – 2024 — **NWO Veni Grant (VI.Veni.202.124)**, Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow's Machine Learning

- PI/Grant holder: Matthijs Vákár.
- My role: Led the work on Iterative CHAD, contributing to the formal methods component of the project. Additionally, established the correctness of dual-numbers-based (forward- and reverse-mode) automatic differentiation for languages with recursive types.

2020 – 2022 — **Marie Skłodowska-Curie Grant**, Principled AD Program Transformations

- PI/Grant holder: Matthijs Vákár.
- My role: Developed the correct extension of reverse-mode automatic differentiation (CHAD) program transformations to expressive total languages.

2011 – 2011 — **Project: Nonlinear Analysis and Applications (CNPq)**, Group Project (University of Brasília), funded by CNPq

- Project title: Nonlinear Analysis and Applications.
- My role: Undergraduate student; led the subproject "Topological Dynamics and Applications in Number Theory" and authored two monographs on the topic.

EARLY-CAREER FELLOWSHIPS AND SCHOLARSHIPS

2020 – 2025 **PhD Fellowship (Promovendus)**, Department of Information and Computing Sciences, Utrecht University, The Netherlands.

2013 – 2017 **PhD Fellowship**, National Council for Scientific and Technological Development (CNPq), Brazil.

2012 – 2013 **Doctoral Research Grant**, Centre for Mathematics, University of Coimbra, Portugal.

2009 – 2012 **Undergraduate and Master's Scholarships**, National Council for Scientific and Technological Development (CNPq) and IMPA, Brazil.

2005 **Selective Pre-University Academic Training**, EPCAR — Brazilian Air Force Preparatory School for Aeronautics, Brazil.

PhD Supervision

PhD STUDENTS

2025 – present **University of Coimbra**, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)

- Student: **Néot Choserot**.
- Research Project: Aspects of Two Dimensional Universal Algebra
- Funding: Centre for Mathematics, University of Coimbra, Portugal.

2026 – present **Utrecht University**, Department of Information and Computing Sciences

- Student: **Diogo Simm**.
- Research Project: Programming language semantics, Probabilistic Programming, DSLs, and Automatic Differentiation.
- Co-supervised with Dr. Matthijs Vákár.
- Funding: ERC Starting Grant *FoRECAST*.

2026 – present **University of Coimbra**, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)

- Student: **Andrea Donati**.
- Research Project: A Galois Theory for Protomodular Objects
- Applied for Funding

FORMER PhD STUDENT

2020 – 2024 **University of Coimbra**, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)

- Student: **Rui Rodrigues de Abreu Fernandes Prezado**.
- Thesis: *Some Aspects of Descent Theory and Applications*.
- Research area: Generalised multicategories and Grothendieck descent theory.
- Co-supervised with Prof. Maria Manuel Clementino.
- Funding: FCT (Fundação para a Ciência e a Tecnologia), Portugal.
- PhD awarded *summa cum laude* (January 2024).

Graduate Supervision

MASTER'S STUDENTS

2026 – 2028

University of Cape Town, South Africa, Master's Programme in Pure Mathematics

- Student: **Kate Birch**.
- Research area: Category theory.
- Co-supervised with Prof. Thomas Meyer.

2024–2025

IMPA, Rio de Janeiro, Brazil, Master's Programme in Pure Mathematics

- Student: **Diogo Simm**.
- Research area: Programming language semantics, automatic differentiation, and category theory.
- Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár.
- Funding: National Council for Scientific and Technological Development CNPq, Brazil.
- Degree awarded with distinction (Grade A).

SHORT-TERM MASTER'S PROJECT SUPERVISION

2025

Utrecht University, The Netherlands, Scientific Methods in Computing Science (Programming Languages)

- Supervision of multiple small-group research projects at Master's level.
- Topics included symbolic and automatic differentiation, program transformations, and categorical semantics.
- Representative projects:
  - *Calculating the Symbolic Derivative using Automatic Differentiation.*
  - *Lambda-Set Specialisation for Haskell.*
  - *Dual-Numbers Automatic Differentiation with Inductive Types.*

Undergraduate Supervision

UNDERGRADUATE THESIS AND CAPSTONE SUPERVISION

2023 — **Utrecht University**, BSc Capstone Project in Computer Science

- Students: **Cérine Touber**, **Oscar Mensink**, **Kasper Peeters**, **Nigel Strachan**, **Joost Laan**, **Jippe Hoogeveen**, **Gust Verhees**, **Nasr Antar**, and **Artur Sałek**.
- Project: Design and implementation of a software system for identifying optimal collaboration matches within organisations, based on interaction data and weighted optimisation criteria.
- Client: Dr. Ioanna Lykourentzou
- Research Areas: Optimisation algorithms, software engineering, and web-based systems.

UNDERGRADUATE SOFTWARE PROJECTS

2021–2022 — **Utrecht University**, Game Technology

- Supervision of nine undergraduate students organised into three project teams.
- Project: Design and development of a game implemented in C#.

Teaching, Curriculum Development, and Pedagogical Leadership

COURSE DESIGN AND CURRICULUM DEVELOPMENT

2026 **University of Coimbra, Portugal**, BSc programme in Mathematics

- **Full design and development of the course *Object-Oriented Programming.***
- Designed the course syllabus, learning outcomes, weekly structure, assessment model, and programming-language orientation.
- Developed the course as a modern introduction to object-oriented programming, connecting software design, abstraction, modularity, and contemporary programming practice.

2026 **University of Coimbra, Portugal**, BSc programme in Mathematics

- **Contribution to the redesign and development of the course *Programming Methods I and II.***
- Contributed to the revision of the course objectives, programme alignment, methodological structure, and articulation with subsequent programming-oriented courses.

SELECTED TEACHING EXPERIENCE

2026 – 2027 **University of Coimbra, Portugal**, BSc program in Mathematics, Computer Engineering and in Artificial Intelligence and Data Science

- **Main Lecturer**.
- Course: *Mathematical Analysis I.*

2026 – 2027 **University of Coimbra, Portugal**, Joint PhD Programme in Mathematics (Coimbra–Porto)

- **Main Lecturer**.
- Course: *Categories in Algebra and Topology.*
- Topics include enriched categories, monads, and generalised multicategories.

2026 – 2027 **University of Coimbra, Portugal**, BSc program in Mathematics, and Computer Engineering

- **Main Lecturer**.
- Course: *Object-Oriented Programming.*

2026 – 2027 **University of Coimbra, Portugal**, BSc program in Mathematics, Computer Engineering and in Artificial Intelligence and Data Science

- **Main Lecturer**.
- Course: *Introduction to Linear Algebra.*

2026 – 2027 **University of Coimbra, Portugal**, BSc program in Mathematics, and Computer Engineering

- **Main Lecturer**.
- Course: *Programming Methods I.*

2025 – 2026 **University of Coimbra, Portugal**, BSc program in Computer Engineering and in Artificial Intelligence and Data Science

- **Main Lecturer**.
- Course: *Mathematical Analysis II.*

2025 – 2026 **University of Porto, Portugal**, Joint PhD Programme in Mathematics (Coimbra–Porto)

- **Main Lecturer**.
- Course: *Categories in Algebra and Topology.*
- Topics include enriched categories, monads, and generalised multicategories.

2020 – 2025 **Utrecht University, The Netherlands**, Department of Information and Computing Sciences

- **Graduate Teaching Assistant (PhD appointment).**
- Functional Programming — tutorials and assessment.
- Logic for Computer Science — tutorials and exercises lessons.
- Graphics for Computer Science (**linear algebra** and applications to Graphics).
- Supervision of BSc Computer Science capstone software projects.
- Coaching and supervision of undergraduate Game Technology projects.
- Recipient of the Utrecht University Distinguished Teaching Award (2023).

2023 – 2024 **Utrecht University, The Netherlands**, Department of Information and Computing Sciences

- **Main Lecturer**, BSc Computer Science.
- Course: *Logic for Computer Science* (co-taught with Dr. Wouter Swierstra).
- Topics included inference systems, natural deduction, semantics, and Hoare logic.

2020 **University of Porto, Portugal**, Joint PhD Programme in Mathematics (Coimbra–Porto)

- **Main Lecturer**.
- Course: *Categories in Algebra and Topology.*
- Topics included enriched categories, monads, and generalised multicategories.

2018 – 2019 **University of Coimbra, Portugal**, Department of Mathematics

- **Invited Assistant Professor**.
- Course: *Topology and Linear Analysis* (co-taught with Prof. Maria Manuel Clementino).

2017 – 2020 **University of Coimbra, Portugal**, Department of Mathematics

- **Instructor, Delfos Project**.
- Advanced mathematics teaching for secondary-school students preparing for national and international mathematics competitions.

2008 – 2012 **University of Brasília, Brazil**, Department of Mathematics

- **Undergraduate Teaching Assistant**.
- Courses included differential and integral calculus, real analysis, linear algebra, and algebra I.

2006 – 2008 **Curso Dínatos, Brasília, Brazil**, Secondary Education

- **Secondary Mathematics Teacher**.
- Mathematics instruction for students preparing for national university entrance examinations.

2005 – 2006 **EPCAR, Brazilian Air Force, Brazil**, Secondary Education

- **Student Teaching Assistant**.
- Mathematics instruction for pre-cadet students.

Invited Lectures and Talks

SELECTED INVITED AND PLENARY LECTURES

UniMont, Milan, Italy, 2026	**CatAlg 2026** , Invited lecture (to be held in July 2026).
University of Messina, Italy, 2026	**2nd Joint Meeting Brazil-Italy in Mathematics**, Invited lecture (to be held in September 2026).
Itaca Fest, Italy 2026	**Itaca Fest 2026**, Invited lecture: A Generalization of the Eilenberg–Moore Factorization.
Lorentz Center, Leiden, The Netherlands, 2022	**New Challenges in Programming Language Semantics**, Invited lecture: *A Categorical Approach to Semantic Logical Relations for Partial Features and Recursive Types.*
University of the Azores, Portugal, 2018	**International Conference on Category Theory 2018**, Plenary lecture: *Aspects of Descent via Bilimits.*

SELECTED TALKS (INVITED AND/OR REFEREED)

Leiden University, The Netherlands, 2024	**109th Peripatetic Seminar on Sheaves and Logic**, The semantic lax descent factorization of a functor.
University of Santiago de Compostela, Spain, 2024	**International Conference in Category Theory 2024**, Doubly-infinitary distributive categories
University of Coimbra, Portugal, 2024	**SUMTOPO 2024, Topology and Logic in Computer Science**, Homomorphic reverse differentiation for partial features
University of Coimbra, Portugal, 2024	**38th Summer Conference on Topology and its Applications**, Free distributive and extensive categories
University of Coimbra, Portugal, 2023	**XIV Portuguese Category Seminar**, Freely generated bicartesian categories and exponentials
Université catholique de Louvain, Belgium, 2023	**International Conference in Category Theory 2023**, Spanning the tale of “Monades et descente”
University of Coimbra, Portugal, 2022	**XIII Portuguese Category Seminar**, CHAD: elementary categorical aspects of automatic differentiation
University of Primorska, Slovenia, 2021	**8th European Congress of Mathematics**, Eilenberg-Moore, Kleisli and descent factorizations.
University of Coimbra, Portugal, 2019	**XII Portuguese Category Seminar**, Lax comma 2-categories.
University of Edinburgh, Scotland, 2019	**International Conference on Category Theory 2019**, Descent and Monadicity.
Stellenbosch University, South Africa, 2019	**Categorical Structures in Algebra, Topology and Logic**, Eilenberg-Moore, Kleisli and descent factorization.
Masaryk University, Brno, Czech Republic, 2018	**103rd Peripatetic Seminar on Sheaves and Logic**, Biadjoint Triangles and Applications.
Universidade de Santiago de Compostela, Spain, 2018	**102nd Peripatetic Seminar on Sheaves and Logic**, Non-canonical isomorphisms.
Université catholique de Louvain, Belgium, 2017	**5th Workshop on Categorical Methods in Non-Abelian Algebra**, Biadjoint Triangles.
University of Coimbra, Portugal, 2017	**XI Portuguese Category Seminar**, Freely generated n-categories and coinserters.

University of Coimbra, Portugal, 2016	**4th Workshop on Categorical Methods in Non-Abelian Algebra**, Bilimits Commutativity and Descent
University of Aveiro, Portugal, 2015	**International Conference on Category Theory 2019**, Kan extensions and descent theory.

WORKSHOP TALKS

LAFI, London, UK	**LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024)**, Homomorphic Reverse Differentiation of Iteration.

(Invited 1-Hour Seminar) TALKS

University of Birmingham, UK, 2026	**Theory of Computation Seminar**, Eilenberg-Moore, Kleisli, and Descent Factorizations
IMPA, Brazil, 2025	**Algebra Seminar**, Biadjoint Triangles and Coherence
University of Cape Town, South Africa, 2025	**Category Theory Seminar**, Bifibrations, Spans, and the Bénabou-Roubaud Theorem
Stellenbosch University, South Africa, 2025	**Category Theory Seminar**, Two-dimensional monad theory, pseudodistributive laws, and doubly-infinitary distributive and extensive categories as pseudoalgebras
University of Cape Town, South Africa, 2025	**Category Theory Seminar**, Two-dimensional monad theory, and free extensive completions
University of Coimbra, Portugal, 2024	**Algebra, Logic and Topology Seminar**, Unraveling the iterative CHAD.
University of Coimbra, Portugal, 2022	**Algebra, Logic and Topology Seminar**, Descent for Split Fibrations.
Utrecht University, Netherlands, 2021	**Software Technology Group Seminar**, Automatic differentiation of expressive total functional languages.
University of Coimbra, Portugal, 2019	**Seminar of the Category Theory Group at the University of Coimbra**, On monadicity and descent.
University of Coimbra, Portugal, 2019	**Algebra, Logic and Topology Seminar**, Functorial Semantics and Descent II.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Functorial Semantics and Descent.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Biadjoint Triangles and Applications.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Towards semi left exactness and Janelidze-Galois within the lax idempotent context.
University of Coimbra, Portugal, 2016	**Algebra, Logic and Topology Seminar**, The deficiency of categories.
University of Coimbra, Portugal, 2015	**Algebra, Logic and Topology Seminar**, Biadjoint triangles, descent and coherence.
Universidade do Porto, Portugal, 2016	**Research Seminar Program**, Kan construction of adjunctions.
University of Coimbra, Portugal, 2016	**Research Seminar Program**, Homotopy Excision.
University of Coimbra, Portugal, 2013	**Joint PhD Program UC-UP Seminar**, Every prespectrum represents a homology theory.

Universidade de Brasília, Brazil, 2010	**Math Seminar**, Topological Dynamics and Applications in Number Theory (in Portuguese).
Universidade de Brasília, Brazil, 2010	**Math Seminar**, Dynamical Proof of the Van der Waerden Theorem (in Portuguese).

Evaluation Committees

DOCTORAL EXAMINATION COMMITTEES

2024 — **PhD Defense Committee**, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Rui Rodrigues de Abreu Fernandes Prezado**.
- Thesis: *Some Aspects of Descent Theory and Applications.*
- Role: Supervisor and examiner.

2023 — **PhD Defense Committee**, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Carlos Miguel Alves Fitas**.
- Thesis: *On Lax Idempotent Monads in Topology.*
- Role: Main examiner and opponent.

DOCTORAL PROJECT EVALUATION

Member of doctoral project evaluation committees within the Joint University of Coimbra–University of Porto PhD Programme in Mathematics (2019), assessing research proposals in category theory and topology, including projects on lax idempotent monads and Grothendieck descent theory. Participated as examiner and supervisor in the formal evaluation of doctoral research plans.

RECRUITMENT

2025 – 2026 — **Hiring and Recruitment Committee**, ERC Starting Grant FoRECAST

- Served as a member and advisor of the recruitment committee for three PhD positions funded by the ERC Starting Grant project *Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy.*

OTHER ACADEMIC EVALUATIONS

Evaluation of Bachelor's Theses, BSc in Artificial Intelligence, Utrecht University, The Netherlands

Academic Service and Governance

GOVERNANCE

2025 – present	**Faculty Member**, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal.
2023 – 2025	**Member, Informatics Research Advisory Committee (IRAC)**, Utrecht University, The Netherlands.
2020	**Faculty Member**, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal.

EDITORIAL AND REFEREEING ACTIVITY

Referee for leading international journals and conferences in mathematics and computer science, including but not limited to: *ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, *Advances in Mathematics*, *Applied Categorical Structures*, *Homology, Homotopy and Applications*, *Journal of Pure and Applied Algebra*, *Mathematical Structures in Computer Science*, *Order*, *Quaestiones Mathematicae*, *Theory and Applications of Categories*, and *Topology and its Applications*.

ORGANISATION OF SCIENTIFIC EVENTS AND SEMINARS

2026 – present	**Organiser, Algebra, Logic and Topology Seminar**, Centre for Mathematics, University of Coimbra, Portugal.
2025 – present	**Organiser, Online Algebra, Logic and Topology Seminar**, Centre for Mathematics, University of Coimbra, Portugal.
2019 – 2023	**Organiser, Algebra, Logic and Topology Seminar**, University of Coimbra, Portugal.
2022	**Organising Committee Member, XIII Portuguese Category Seminar**, University of Coimbra, Portugal.
2019	**Organising Committee Member, XII Portuguese Category Seminar**, University of Coimbra, Portugal.
2018	**Organiser, Workshop on Algebra, Logic and Topology**, University of Coimbra, Portugal.
2018	**Organiser, Workshop: Logic for Children**, UNILOG 2018, Vichy, France.

OUTREACH AND PUBLIC ENGAGEMENT

2026 – present — **CLASS Seminar**, University of Coimbra, University of Cape Town, and Utrecht University

- I lead and organize an international seminar (called CLASS – Category, Logic, Algebra and Semantics Seminar) for graduate students working on categorical structures. The seminar brings together participants from Computer Science and Pure Mathematics, and multiple countries, including South Africa, United Kingdom, the Netherlands, Portugal, and Germany, fostering collaboration and the exchange of ideas across them.

2017 – 2020 — **Delfos Project**, Department of Mathematics, University of Coimbra, Portugal

- Contributor to the Delfos Project, a mathematics education programme for secondary-school students preparing for international mathematics competitions.
- Served as Portuguese representative to the International Mathematical Olympiad (IMO) in 2020.

2008 – 2012 **Scientific Initiation (CNPq)**, Brazil

- Undergraduate member of the research group *Teoria de Lie e Dinâmica*.
- Contributed to the outreach blog `https://liedinamica.wordpress.com/`.
- Authored three educational monographs in Portuguese on Lie groups, topology for dynamical systems, and applications of topological dynamics to number theory.
- Published an expository article in *Revista Matemática Universitária* (Portuguese).

2011 – 2012 **Brazilian Society for Scientific Progress (SBPC)**, Brazil

- Presented multiple popular-science posters in mathematics at national SBPC conferences.
- Topics included game theory, homotopy theory, category theory, topology, and dynamical systems.
- All contributions were published in the conference annals (in Portuguese).

LANGUAGES

Portuguese and English (and studying Dutch)